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RESEARCH ARTICLE

Enhancing Cybersecurity of a Hydroelectric Power Plant Through Digital Twin Modeling and Explainable AI

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ABSTRACT Hydroelectric power plants (HEPPs) are vital components of the renewable energy infrastructure, making their operational security and efficiency critical. HEPPs face increasing vulnerability to cyber threats, which can disrupt operations and compromise energy production. This study investigates the integration of Digital Twin (DT) technology with Explainable Artificial Intelligence (XAI) to improve cybersecurity and anomaly detection in a HEPP located in Türkiye. The DT model simulates key operational parameters, such as water flow, mechanical power, and turbine speed, enabling real-time monitoring, optimization, and secure cybersecurity testing. Simulated cyberattacks on the DT have revealed vulnerabilities in the Modbus protocol, while SHapley Additive exPlanations (SHAP) analysis, an XAI technique, clarifies the influence of operational parameters on anomaly detection outcomes. Gravity with a SHAP value of 0.0001 and water density with a SHAP value of 0.0018 have been identified as the least influential parameters, suggesting limited variability or impact on the detected anomalies. These findings enable the prioritization of critical variables while reducing unnecessary monitoring efforts. The proposed integration improves the accuracy of anomaly detection, enables precise vulnerability identification, and mitigates operational risks without impacting real-world systems. The results demonstrate the effectiveness of the model in detecting anomalies and strengthening the resilience of the system against cyber threats. This approach also provides actionable insights to optimize operational processes, ensuring secure and efficient energy production. The study highlights the transformative potential of DT-XAI integration in advancing the sustainability, security, and resilience of critical energy infrastructures.

INDEX TERMS Cyber security, digital twins, explainable artificial intelligence, hydroelectric power plant.

I. INTRODUCTION

HEPPs are among the world's largest and oldest renewable energy sources, and they produce energy without harming the environment by converting the potential energy of water into electrical energy. They provide approximately 29 percent of the electricity production in the world and 31 percent in Türkiye. The installed capacity and production of HEPPs depend on factors such as water flow and net decline. For example, in Türkiye, the installed capacity of HEPPs in 2022 has been 31,000 MW and its production has been 66 GWh. In addition, with the newly added installed capacity of 558 MW, Türkiye has become the 9th country in the world to increase its capacity in 2022 [1]. In this respect, it is observed from the statistics that the importance of the energy produced from HEPPs in Türkiye is increasing day by day.

HEPPs have been observed to provide a significant amount of electricity production in Türkiye and around the world. In this regard, any functional failure, interruption, or halt in the production of these power plants can have serious consequences. Depending on the production capacity, ensuring the business continuity of these power plants is of vital importance in terms of public order, national economy,

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and security. A system of such importance can naturally be targeted by advanced threat actors, and attacks can be carried out on these systems using different cyber attack methods. Cyberattacks can disrupt HEPP control systems, reduce operational performance, cause data loss or leakage, and cause environmental or human-related damage [2].

Cybersecurity tests are performed to measure the resistance of HEPPs against cyberattacks and to identify their vulnerabilities. They can be performed in real power plant environments or in virtual environments. Tests performed on real power plant can provide more realistic results, but they can also be riskier and costlier when performed in a production environment [3]. If an error occurs during a test performed in a real environment, it can stop the power plant from running or cause damage. Tests performed in virtual environments can be safer and cheaper, but may not reflect all the dynamics of the real environment. During a test performed in a virtual environment, the full impact of an attack may not be seen, or some attack scenarios may be overlooked.

In order to effectively manage these facilities and optimize HEPP's performance, the concept of digital twins offers an important perspective today. A digital twin is an information technology tool that creates a virtual model of a physical system and can analyze data in real time on this model [4]. The process of designing a digital twin in HEPPs has the potential to increase efficiency in operation and maintenance processes by accurately reflecting the complex structure of the facility.

Digital twins offer a wide range of applications to improve the performance of HEPPs and optimize operating costs. These systems simulate plant energy production, water management, and equipment status in a virtual environment, providing operators with a more reliable and effective decision-making process [5]. In addition, digital twins increase plant reliability by supporting processes such as developing maintenance strategies and creating intervention plans based on predefined scenarios.

HEPPs are among critical infrastructures that play an important role in energy production and therefore have a high potential to be targeted. Ensuring the operational safety of power plants is of great importance not only for the uninterrupted continuation of energy production processes, but also for preventing environmental and economic damage. In this context, digital twin technology offers a powerful tool for monitoring, analyzing, and optimizing the performance of HEPPs.

A digital twin can simulate the behavior of a physical asset or system by feeding it with real-time data and can be used to monitor, analyze, optimize, and improve the performance of a physical asset or system. It can also provide a suitable platform for cybersecurity testing. Cybersecurity tests performed on a digital twin can provide results close to the real environment and can be performed without damaging or disrupting the real environment. For example, during a test performed on a digital twin, it can be seen how the impact of an attack is reflected on all components of the plant, or the plant's cyber resilience can be measured by trying different attack scenarios.

When studies on digital twins are examined, it is seen that they are designed and implemented for different purposes in many different sectors and infrastructures. The digital twin market has been growing rapidly in recent years and is expected to reach 52.4 billion USD from 11.12 billion USD by 2030 [1]. Factors contributing to the growth of the digital twin market include the impact of the COVID-19 pandemic, the spread of IoT technology, the development of artificial intelligence and cloud computing, industrial digitalization and the need for data security. The largest areas of use of the digital twin market include automotive and transportation, manufacturing, energy and utilities, telecommunications and housing sectors.

Within the scope of this article, a digital twin model of an HEPP with a production capacity of 1.2 MW operating in Türkiye has been created. A cybersecurity test has been performed on this model by performing data manipulation on the nozzle aperture, and the results have been analyzed. Then, in the final stage of the study, anomaly detection was performed with XAI techniques and deviations in the operational processes of the power plant were analyzed in detail. This method not only provided anomaly detection, but also provided important information on which parameters should be carefully monitored.

This paper provides several significant contributions to the field of HEPP cybersecurity and optimization, addressing critical aspects of integrating Digital Twin technology and Explainable AI into energy systems:

- Development of a DT model for a HEPP to enable real-time monitoring, performance optimization, and secure cybersecurity testing.
- Integration of XAI techniques, specifically SHAP analysis, to enhance the transparency and reliability of anomaly detection processes.
- Implementation of a risk-free environment for simulating cyberattacks on the HEPP, revealing vulnerabilities in the Modbus TCP protocol, and enabling proactive mitigation strategies.
- Detailed evaluation of operational benefits, including improved anomaly detection, enhanced security, and optimized system performance.
- Proposition of a robust and scalable approach to strengthen the cybersecurity and sustainability of critical energy infrastructures.

The structure of this paper is organized as follows: Section II reviews related works, highlighting advancements and gaps in integration of DT and XAI technologies for cybersecurity of critical infrastructure. Section III introduces the Digital Twin model for the HEPP, describing its architecture, operational principles, and integration with real-time data systems. Section IV explains the developed cybersecurity framework, including the methodology for simulating attacks and the use of XAI techniques for

anomaly detection and explainability. Section V presents experimental results, focusing on the performance of DT-XAI integration to identify vulnerabilities, optimize operations, and mitigate cyber risks. Section VI concludes the paper by summarizing key findings, discussing the limitations of the proposed approach, and suggesting future research directions.

II. RELATED WORKS

Digital twin technology has become a powerful tool for the expansion, diversity, and management of options, offering many innovations in product change in recent years. Virtually merging international systems, this technology is used in a wide range of application areas such as real-time monitoring, predictive maintenance, simulation, and analysis. The contribution of these powerful digital twins to critical areas such as energy systems, industrial applications, smart cities, and cybersecurity is unique. Publications in the literature provide access to detailed expanded categories to reveal the areas of use and methods of digital twin technology. This assessment provides an important roadmap for the current and scalable nature of digital twins and for better storage of this technology.

A. PROCESS OF REVIEW OF STUDIES

The publications included in these studies have been categorized in order to understand the contribution of digital twin technology to different application areas. During the review process, each publication has been evaluated in terms of the problem addressed, the methods used, and the contributions it provides. The following criteria have been taken into account in this evaluation process:

- **Application Area:** The industry or systems the studies focus on, such as power systems, renewable energy, or cybersecurity.
- **Used Methods:** Technical approaches used in studies, such as data analytics, artificial intelligence, IoT integration.
- **Contributions and Implications:** Practical or theoretical innovations offered by studies, such as increasing energy efficiency or ensuring system security.

B. RESEARCH TOPICS AND FOCUS AREAS

In these studies, it is seen that digital twin technology is examined from different aspects. The prominent research topics are as follows:

- **The Role of Digital Twins in Power Systems:** Increasing operational efficiency in power systems, ensuring uninterrupted energy distribution and optimizing predictive maintenance of systems.
- **Optimization of Renewable Energy Systems:** Monitoring and optimizing the production processes of renewable energy sources such as wind turbines and HEPPs.
- **Cybersecurity Solutions:** The ability of IoT-based digital twins to develop security measures against cyber attacks.
- **Smart Grid Management:** Systems that optimize energy consumption and production, integrate renewable energy sources, and ensure network security.

C. METHODS USED

In the publications, the following methods are noteworthy:

- **Data Analytics and Artificial Intelligence:** Big data processing, machine learning and deep learning techniques have been used in system optimization and predictive maintenance.
- **IoT and Real-Time Monitoring:** Real-time monitoring of systems has been achieved by integrating data from IoT devices into digital twin models.
- **Simulation and Modeling:** Testing and improving operational processes by creating virtual prototypes.
- **Cybersecurity and Threat Analysis:** The ability of digital twins to identify vulnerabilities and develop protective measures by simulating potential threats.

D. CONTRIBUTIONS OF THE STUDIES

These publications shed light on the applications of digital twin technology in different areas and make the following important contributions:

- **Operational Efficiency:** Offers innovative approaches that increase system performance in many sectors from energy production to transportation.
- **Sustainability:** Reduces environmental impacts by ensuring more efficient use of renewable energy resources.
- **Security:** Provides cybersecurity in complex systems such as IoT-based infrastructures and smart cities.
- **Education and Simulation:** Provides safe learning environments that simulate real-world applications in the field of education.

Table 1 provides a comprehensive summary of the contributions of studies on digital twin technology across various domains, highlighting the methodologies employed and the core research areas explored. The table categorizes existing literature into specific fields such as power systems, renewable energy, smart grids, industrial sectors, cybersecurity, transportation, smart cities, education, and anomaly detection with XAI. Each category is supported by a carefully selected set of references drawn from relevant studies, offering a detailed overview of the latest advances and emerging trends in digital twin applications. In this regard, the table serves as a valuable resource for researchers and practitioners seeking to understand the multifaceted applications of digital twin technology in different domains. It offers insights into the key contributions, challenges, and future directions of digital twins, emphasizing their transformative potential to drive innovation and operational excellence in various industries.

TABLE 1. Digital twin studies in literature.

Category	References	Explanations and Contributions
Power Systems	[4], [6], [7], [8], [9], [10]	These studies address the application of digital twins in power systems to improve operational efficiency, system optimization, and predictive maintenance. The works also discuss challenges in implementing digital twins, such as data integration, cybersecurity, and IoT connectivity. Notable contributions include incident prediction models, IoT-based architectures, and frameworks for future applications.
Renewable Energy	[11], [12], [13], [14], [15]	Focuses on optimizing energy production processes for renewable sources like wind and hydroelectric power. Contributions include monitoring turbine wear, predicting maintenance needs, optimizing water resource use, and integrating renewable sources into energy grids. These models improve energy efficiency, sustainability, and cost management.
Smart Grids	[16], [17], [18], [19], [20], [21], [22], [23], [24], [25]	Digital twins in smart grids enhance energy distribution, optimize consumption, and ensure resilience. Key contributions involve fault detection, cybersecurity testing, dynamic energy management, and integration of renewable energy. These systems enable real-time monitoring, predictive maintenance, and operational efficiency while supporting decarbonization efforts.
Industrial Sectors	[26], [27], [28], [29], [30], [31], [32], [33], [34], [35]	Industrial uses include SCADA integration, factory automation, and energy management. Digital twins enable detailed simulations, predictive analysis, and operational optimization in manufacturing and energy-intensive industries. Specific contributions include the integration of BIM and SCADA, and the development of hybrid digital twin systems.
Cyber Security	[36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48]	Explores digital twins for proactive cybersecurity measures across industries. Contributions include threat simulation, real-time anomaly detection, and the development of cyberresilient frameworks. Applications extend to smart grids, water systems, and industrial control systems, focusing on early detection of vulnerabilities and enhancing organizational preparedness against cyber threats.
Transportation Systems	[39], [49], [50]	Transportation systems benefit from real-time monitoring and predictive analysis provided by digital twins. Contributions focus on airport traffic control, railway power management, and maritime operations. These systems enhance safety, efficiency, and operational cost reduction
Smart Cities	[51], [52], [53], [35], [54], [14], [55], [56], [57], [58], [30], [59], [60], [61], [62], [63]	Smart city applications include traffic flow optimization, energy management, and resource sustainability. Contributions highlight the integration of renewable energy, waste management optimization, and urban infrastructure simulations. These systems also address cybersecurity challenges and enable better decision-making for city planners.
Education Sectors	[64], [65]	Focuses on using digital twins for immersive and practical learning experiences. Applications include CPS, SCADA systems, and specialized training programs for engineering and medical education. These systems allow safe, controlled simulations to enhance skill development
Anomaly Detection and XAI	[66], [67], [68], [69], [70], [71], [72]	Studies develop methods to make AI-driven anomaly detection more transparent and interpretable. Contributions include using SHAP and LIME for model explainability, counterfactual reasoning for high-dimensional data, and localization of anomalies in IoT systems. These advances enhance trust and reliability in AI systems.

XAI has been used in this study to address the critical need for interpretability in anomaly detection models, particularly in complex systems such as HEPPs. Traditional black-box machine learning models lack transparency, which hinders their practical applicability in safety-critical environments. The SHAP technique has been specifically chosen because of its ability to provide consistent and interpretable explanations for model predictions, enabling domain experts to understand the influence of individual parameters on detected anomalies. This level of transparency is essential for identifying key operational factors and improving the trustworthiness of the anomaly detection process.

A significant contribution of this study is the implementation of a risk-free simulation environment within the digital twin model for cybersecurity testing. This approach allows for the identification of vulnerabilities, such as those in the Modbus TCP protocol, and facilitates the development of proactive mitigation strategies without affecting real-world systems.

A key contribution is the use of SHAP in this study is to analyze the impact of critical parameters in HEPP operations and determine which variables contribute the most or least to system performance. In this way, the effects of parameters in anomaly detection models have been evaluated transparently and it has been determined which variables operators should focus on first. In addition, confidence in decision processes has been increased and user confidence in model outputs has been ensured thanks to SHAP analysis.

The study also optimizes energy production by using the DT model for real-time monitoring and adjustment of critical parameters such as water flow and turbine speed. In addition, it proposes a scalable framework that can be applied to other energy infrastructures, supporting both sustainability and improved resilience.

By combining DT and XAI, the study bridges the gaps in transparency, operational efficiency, and cybersecurity, offering practical solutions to strengthen the security and sustainability of critical energy systems.

This approach not only increased anomaly detection and system security, but also addressed a significant gap in the existing literature by providing operational insights to optimize system performance. The explanations provided with the SHAP technique created an innovative framework that strengthened decision support processes in critical energy infrastructures, such as HEPPs.

III. DIGITAL TWIN OF THE HEPP

In this study, in order to better understand the digital twin model designed by feeding real-time data from the field, it is necessary first to understand the current HEPP infrastructure. In this respect, first, the HEPP structure and architecture from which the digital twin model has been derived will be examined within the scope of the study, and then how the digital twin model has been designed and developed will be explained.

A. HEPP SYSTEM ARCHITECTURE

The system architecture of a HEPP has a complex and integrated network structure. This type of system consists of various components and subsystems in order to optimize both electricity production and increase safety and efficiency. In general, this architecture consists of a central control room (SCADA), generator protection relays, regulators, reservoir control systems, and various protection relays.

The SCADA (Supervisory Control and Data Acquisition) control room is the central management center of the HEPP. This control room monitors and manages all operations of the plant. The SCADA system collects data from field devices and processes these data at a central point. This allows operators to monitor the performance of the plant in real time and intervene when necessary [73]. The SCADA system is also used to analyze and report data.

The regulator and reservoir control systems regulate the water flow and storage of the HEPP. These systems control water levels and flow rates, ensuring that electricity production is continuous and balanced. The regulator directs water to the turbines and controls the turbine speed. The reservoir control system manages sudden floods and drought conditions by keeping the water level within a certain range [74].

Generator protection relays play a critical role in protecting the generators used in HEPPs. These relays detect overcurrent, overvoltage, and other abnormal conditions that may occur in the generators and prevent damage by disabling the generators when these conditions occur [75]. Generator protection relays ensure safe and uninterrupted power plant operation.

Transformer and line protection relays also play an important role in the safety of the HEPP. Transformer protection relays protect transformers from overload, short circuit, and other abnormal conditions. Line protection relays detect faults that may occur in the power lines and prevent these faults from spreading. These relays ensure the safe and efficient operation of the power plant's electrical distribution system [76].

The excitation system is a system that regulates the magnetic field of the generator. This system plays a critical role in controlling the electric current produced by the generator. The excitation system ensures the efficient operation of the generator and increases the quality of electricity production [77]. It also protects the generator from overload and other abnormal conditions.

Finally, turbine control systems and related units control the main energy production units of the HEPP. These systems regulate the speed and power of the turbines and ensure that the turbines operate efficiently [78]. Turbine control systems increase the overall performance of the plant and ensure a constant balance in energy production.

This system architecture consists of a series of integrated components that ensure the safe, efficient, and sustainable operation of the HEPP. Each component performs a specific function and contributes to overall system performance. This structure allows the complex operations of the HEPP to be effectively managed and optimized.

The components used in the HEPP and the general system architecture are shown in Figure 1.

B. NETWORK OF THE HEPP AND DIGITAL TWIN

The network topology of a HEPP is critical to ensuring operational efficiency and maintaining the integrity of the system. The basic components of this topology consist of the power plant equipment, router, digital twin server, real-time data systems, and digital twin data systems. These components are connected to each other through the Internet and local networks.

The main hardware components of the power plant are the devices that provide energy production and distribution. These devices consist of turbines, generators, and other auxiliary equipment that is used to generate electricity in the HEPP. This equipment is connected to the network via a router, which transmits the power plant's data to the digital twin server over the Internet.

The router is a central component of the HEPP's network topology. The router receives data from the power plant equipment and sends it to the digital twin server over the Internet.

The router manages data traffic and ensures that data is transmitted securely. In addition, the router provides communication between the power plant network and the Internet and ensures that the data reach the correct destination.

The digital twin server creates a virtual copy of the HEPP, and this virtual copy is constantly updated with real-time data. The digital twin server collects and processes data using protocols and tools such as Modbus and Elastic. Modbus is a communication protocol that is used to collect data from power plant devices [79]. Elastic is a data processing platform that is used to store and analyze large data sets [80].

The digital twin server processes and stores real data and digital twin data separately. Although real data reflect real-time operations of the HEPP, the digital twin data represent the virtual model of the plant. These two data sets are used to monitor, analyze, and optimize the performance of the plant. The digital twin server integrates with other systems using API (Application Programming Interface) and enables data exchange.

Real-time data systems collect and analyze the operational data of the HEPP. These systems process data from power

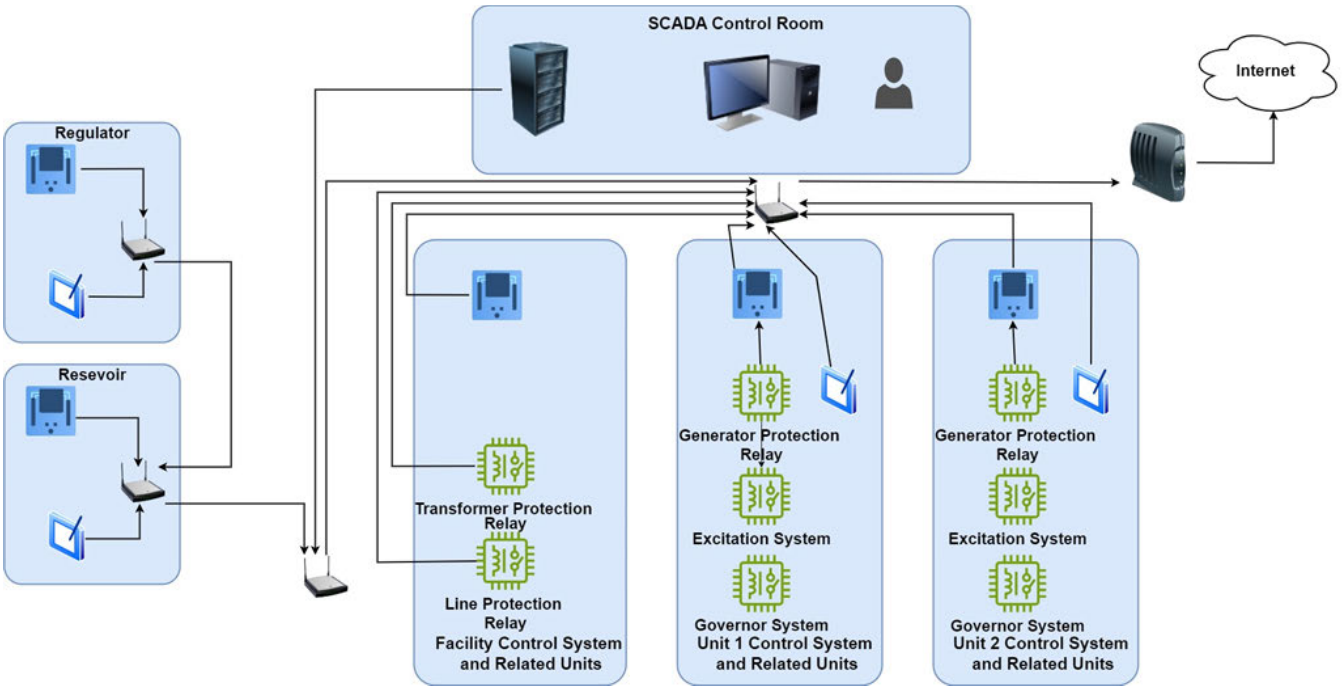


FIGURE 1. HEPP system architecture.

plant devices and facilitate decision-making by operators. Real-time data systems monitor the performance of the power plant and alert operators in case of any abnormalities or faults. These systems are critical to increasing the efficiency of the plant and minimizing operational interruptions.

Digital twin data systems create a virtual model of the plant and ensure that this model is constantly updated. Digital twin data systems simulate power plant operations and make recommendations to optimize power plant performance through these simulations. Digital twin data systems allow power plant operators to test future operational scenarios and assess potential risks.

This network topology ensures that HEPP operates efficiently and safely. Each component performs a specific function and contributes to overall system performance. This structure allows plant operations to be managed and optimized effectively, thus increasing the sustainability and security of energy production.

Communication infrastructure and topology of the HEPP is shown in Figure 2.

C. MODELING DIGITAL TWIN

The Digital Twin system developed for the HEPP integrates various modules to simulate and analyze real-time operational data. The system operates in a structured workflow, starting with data acquisition and proceeding through data transformation, computational modeling, and visualization.

The digital twin system created for the HEPP includes the processes of collecting, processing, and analyzing the operational data of the plant. This system works in an integrated manner with various components and data flows.

Data collection from the plant site is carried out using the Modbus protocol. Modbus is a communication protocol widely used in industrial automation systems and provides an effective method to read data from plant devices. Various sensors and devices in the plant site collect operational data via Modbus and the data are transmitted to a central data collection unit.

The collected data are recorded by writing to the Modbus server. During this process, the accuracy and integrity of the data are checked and the necessary preliminary processes are applied. The data written to the Modbus server is then prepared to be used for the digital twin system. This stage is critical for correct data recording and usability in the following processes.

In the digital twin system, converting the data to JSON format is an important step. JSON provides a lightweight and easy-to-understand format for data exchange. Data received from the Modbus server are converted to JSON format, providing a more flexible and portable data structure. This conversion allows the data to be easily processed by the digital twin system and other analysis tools, but introduces additional computational overhead, which can impact system performance during high-frequency operations.

The data converted to JSON format is sent to Elasticsearch. Elasticsearch is a powerful search and analysis engine that is used to store, search, and analyze large data sets. Sending digital twin data to Elasticsearch ensures that the data is quickly accessible and ready for analytical processes. Elasticsearch offers high performance and flexibility in storing and managing data.

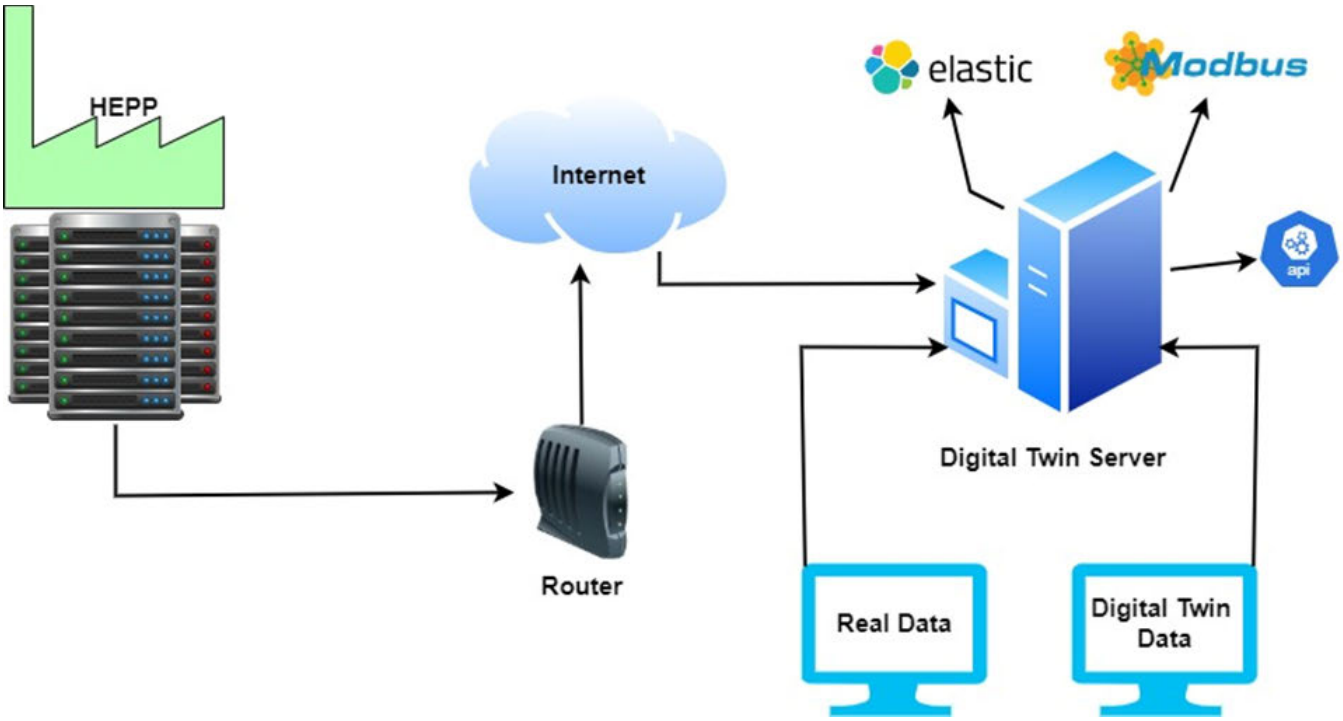


FIGURE 2. Communication topology of the HEPP.

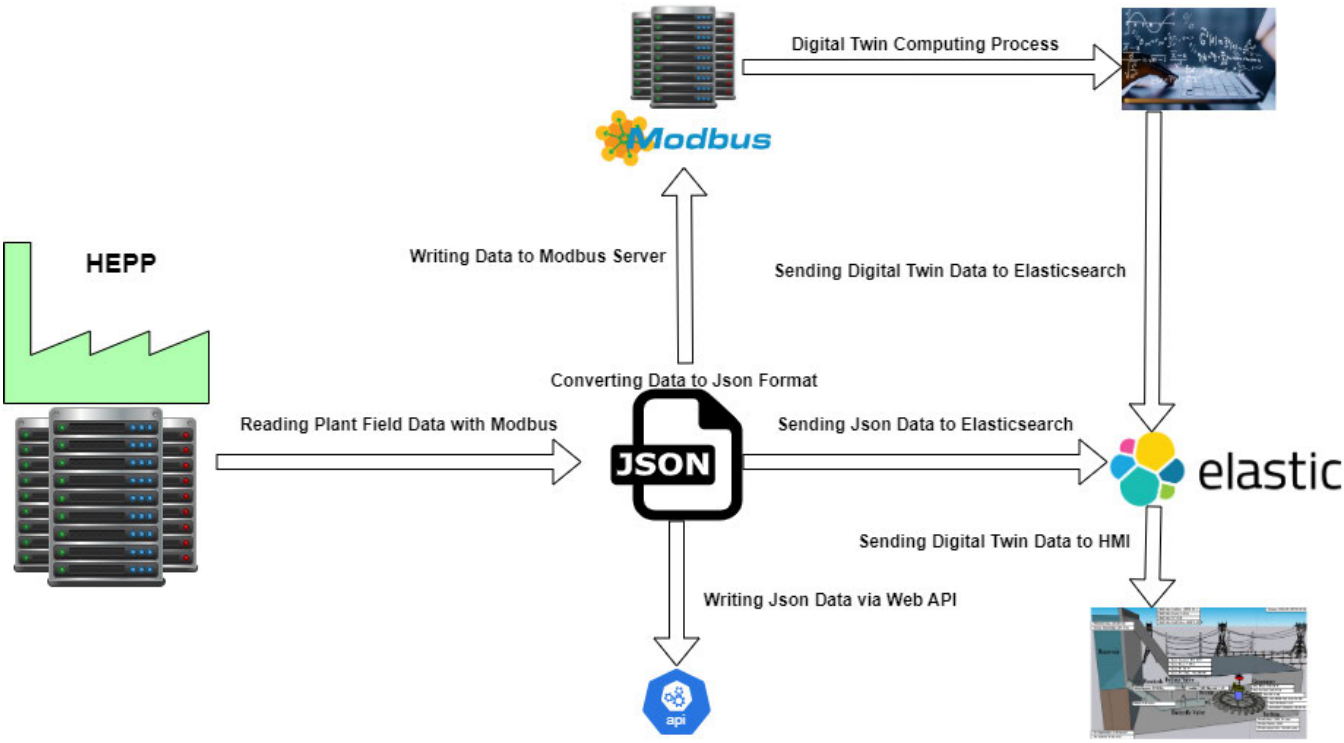


FIGURE 3. Digital twin flow diagram of the HEPP.

The digital twin calculation process involves creating a virtual copy of the physical system and performing simulations and analyses on this copy. Using the collected data and the digital twin model, various scenarios such as plant performance, efficiency, and possible failure situations are analyzed. This process provides a scientific basis for plant

operators in making future operational decisions. However, its calculation intensity and the need for precise calibration to mirror real-world conditions remain challenges.

The digital twin data are also transmitted to the (Human-Machine Interface) HMI system. HMI is an interface that allows operators to monitor and control plant operations. The integration of digital twin data into HMI makes it easier for operators to monitor plant performance in real time and make necessary interventions. This integration enables operators to make more informed and timely decisions.

Another important component of the digital twin system is the web APIs. Web APIs are interfaces that enable data exchange between different systems. JSON data is written to the digital twin system via web APIs, and the data is shared with other analysis and monitoring systems. Web APIs increase the flexibility of the digital twin system and enable various integration scenarios.

As a result, HEPP's digital twin system manages the processes of collecting, processing, analyzing, and monitoring the operational data of the plant in an integrated manner. This system has been developed to increase the efficiency of the plant, reduce operational risks, and optimize decision-making processes. Digital twin technology offers significant advantages in complex engineering systems such as HEPPs and ensures the sustainability and security of such systems.

The operation flow diagram of the digital twin system of the HEPP is shown in Figure 3.

D. MATHEMATICAL MODEL OF THE DIGITAL TWIN FOR THE HEPP

Data collected in the HEPP are collected from various units and addresses and integrated into the digital twin system. This process is of critical importance for monitoring, analyzing, and optimizing the performance of the power plant.

Each data variable is collected from a specific HEPP address and written to the relevant addresses of the digital twin system. In addition, the equations used for digital twin calculations are also part of this integration [81]. The mathematical formulations utilized in the digital twin computations are presented in the following equations, offering a detailed representation of the operational dynamics of the HEPP. Furthermore, the source of each data variable, including the specific Modbus register addresses from which they are acquired and the corresponding digital twin system registers where they are stored, is systematically detailed in Table 2. This comprehensive table not only provides information on the data mapping process, but also ensures precise synchronization between the physical and virtual environments, facilitating real-time monitoring, analysis, and optimization of plant operations. Furthermore, the functional relationships between variables, and equations, as illustrated in Figure 4, are elaborated within the equation explanations, clarifying the role of each function in the overall system performance.

1) WATER HEIGHT (RESERVOIR POOL MODULE)

$$H = B_i \cdot g \tag{1}$$

where H represents the water height, B_i is the inlet pressure, and g denotes the gravitational acceleration. The function f_1 represents the relationship between inlet pressure and gravity to determine water height within the reservoir. The accurate calculation of water height is essential to determine the available potential energy in the reservoir, which directly impacts the efficiency of the HEPP.

2) WATER FLOW (RESERVOIR POOL MODULE)

$$Q = (nz1 + nz2) \cdot P_{flow}/P_{pos} \tag{2}$$

where Q is the water flow, $nz1$ and $nz2$ are the nozzle apertures, P_{flow} is the project flow, and P_{pos} represents the project position. The function f_2 calculates the total water flow based on nozzle apertures and project parameters. Understanding water flow dynamics allows better control of power generation by optimizing nozzle apertures and ensuring efficient use of the available water resources.

3) MECHANICAL POWER (RESERVOIR POOL MODULE)

$$Pm = g \cdot p \cdot H \cdot Q \cdot n \tag{3}$$

where P_m is the mechanical power, p is the water density, and n is the efficiency. Function f_3 determines the mechanical power generated from water flow. Mechanical power is a key indicator of the plant's operational efficiency, and its accurate calculation helps in predicting energy output and optimizing performance.

4) GENERATOR ACTIVE POWER (GENERATOR MODULE)

$$Gen_{pa} = 1.73 \cdot Gen_v \cdot Gen_i \cdot Gen_{pf}/1000 \tag{4}$$

where Gen_{pa} is the generator active power, Gen_v represents generator voltage, Gen_i is the generator current, and Gen_{pf} denotes the generator power factor. The function f_4 quantifies the active power output of the generator. Active power of the generator directly affects the electricity supplied to the grid, making it crucial to monitor and optimize for stable operation.

5) GENERATOR REACTIVE POWER (GENERATOR MODULE)

$$Gen_{rp} = 1.73 \cdot Gen_v \cdot Gen_i \cdot (1 - Gen_{pf})/100 \tag{5}$$

where Gen_{rp} is the reactive power of the generator and the function f_5 quantifies the reactive power output of the generator. Reactive power management is essential for maintaining voltage levels within the system and ensuring the reliable operation of electrical equipment.

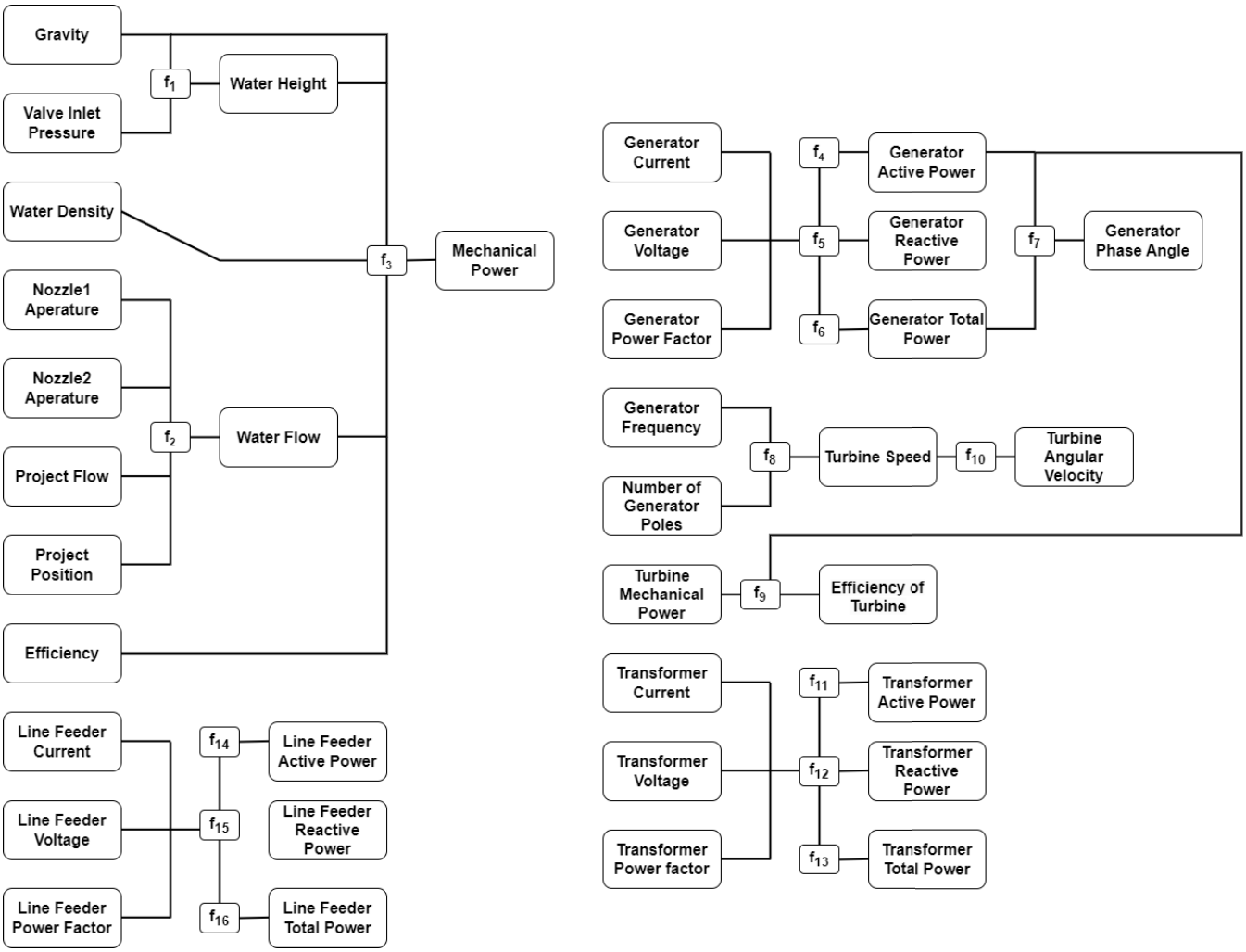


FIGURE 4. Digital twin model functional diagram of the HEPP.

6) GENERATOR TOTAL POWER (GENERATOR MODULE)

$$Gen_{sp} = 1.73 \cdot Gen_v \cdot Gen_i \tag{6}$$

where Gen_{sp} is the generator total power, and function f_6 quantifies the total power output of the generator. Total power calculation provides a comprehensive understanding of the performance of the generator by considering both active and reactive power components.

7) GENERATOR PHASE ANGLE (GENERATOR MODULE)

$$Gen_{\phi} = \arccos(Gen_{pa}/Gen_{sp}) \tag{7}$$

where Gen_{ϕ} is the generator phase angle, and function f_7 determines the phase angle of the generator system. The generator phase angle is critical for synchronizing with the grid, ensuring stable and efficient power transmission.

8) TURBINE SPEED (TURBINE MODULE)

$$Tb_s = 120 \cdot Gen_f / Gen_{pole} \tag{8}$$

where Tb_s is the turbine speed, Gen_f represents generator frequency, and Gen_{pole} denotes the number of generator poles. The tunction f_8 estimates the rotational speed of the turbine. Turbine speed plays a vital role in achieving optimal generator operation and maintaining the desired frequency of the generated electricity.

9) TURBINE EFFICIENCY (TURBINE MODULE)

$$Tb_n = Gen_{pa}/Tb_{pm} \tag{9}$$

where Tb_n is the turbine efficiency, Gen_{pa} represents generator active power, and Tb_{pm} is the turbine mechanical power. The function f_9 evaluates the efficiency of the turbine system. Maximizing turbine efficiency contributes to reducing energy losses and improving the overall sustainability of the HEPP.

10) TURBINE ANGULAR VELOCITY (TURBINE MODULE)

$$Tb_w = Tb_s \cdot 2 \cdot \pi/60 \tag{10}$$

TABLE 2. Mapping of operational parameters in the digital twin model.

Unit	Variable	Explanation	HEPP Address	Digital Twin Address
Reservoir Pool	g	Gravity	4448	20
	p	Water Density	4836	18
	H	Water Height	4672	4
	Q	Water Flow	6874	2
	n	Efficiency	6892	22
	P_m	Mechanical Power	6876	0
	$nz1$	Nozzle 1 Aperture	4632	24
	$nz2$	Nozzle 2 Aperture	4634	26
	P_{flow}	Project Flow	6004	64
	P_{pos}	Project Position	6002	66
	B_i	Inlet Pressure	4662	40
Generator	Gen_i	Generator Current	4608	36
	Gen_v	Generator Voltage	4620	34
	Gen_{pf}	Generator Power Factor	4610	42
	Gen_{pa}	Generator Active Power	6568	12
	Gen_{rp}	Generator Reactive Power	6536	14
	Gen_{sp}	Generator Total Power	6554	70
	Gen_{ϕ}	Generator Phase Angle	4612	48
	Gen_f	Generator Frequency	4606	38
	Gen_{pole}	Pole Number of Generator	6718	68
Turbine	Tb_s	Turbine Speed	6742	10
	Tb_n	Turbine Efficiency	6892	72
	Tb_w	Turbine Angular Speed	6724	28
	Tb_{pm}	Turbine Mechanical Power	6876	62
Transformer	Trf_v	Transformer Voltage	4602	50
	Trf_i	Transformer Current	6786	52
	Trf_{pf}	Transformer Power Factor	4592	54
	Trf_{pa}	Transformer Active Power	4596	74
	Trf_{rp}	Transformer Reactive Power	4598	76
	Trf_{sp}	Transformer Total Power	4600	78
Line Feeder	Hf_v	Line Feeder Voltage	6738	56
	Hf_i	Line Feeder Current	6730	58
	Hf_{pf}	Line Feeder Power Factor	6712	60
	Hf_{pa}	Line Feeder Active Power	6740	80
	Hf_{rp}	Line Feeder Reactive Power	6706	82
	Hf_{sp}	Line Feeder Total Power	6708	84

where Tb_w is the turbine angular velocity, and Tb_s represents turbine speed. The function f_{10} determines the rotational speed of the turbine in radians per second. Turbine angular velocity is a critical parameter for ensuring mechanical stability and preventing excessive wear and tear on the system.

11) TRANSFORMER ACTIVE POWER (TRANSFORMER MODULE)

$$Trf_{pa} = 1.73 \cdot Trf_v \cdot Trf_i \cdot Trf_{pf}/100 \tag{11}$$

where Trf_{pa} is the transformer active power, Trf_v represents transformer voltage, Trf_i is the transformer current, and Trf_{pf} denotes the transformer power factor. The function f_{11} quantifies the active power delivered by the transformer. Active power measurement from transformers is crucial to determine effective power transfer to the electrical grid while minimizing losses.

12) TRANSFORMER REACTIVE POWER (TRANSFORMER MODULE)

$$Trf_{rp} = 1.73 \cdot Trf_v \cdot Trf_i \cdot (1 - Trf_{pf})/100 \tag{12}$$

where Trf_{rp} is the transformer reactive power, and the function f_{12} quantifies the active power delivered by the transformer. Effective reactive power control in transformers ensures power factor correction and enhances system stability.

13) TRANSFORMER TOTAL POWER (TRANSFORMER MODULE)

$$Trf_{sp} = 1.73 \cdot Trf_v \cdot Trf_i \tag{13}$$

where Trf_{sp} is the transformer total power, and the function f_{13} quantifies the active power delivered by the transformer. Calculating total transformer power helps in capacity planning and efficient utilization of electrical energy.

14) LINE FEEDER ACTIVE POWER (LINE FEEDER MODULE)

$$Hf_{pa} = 1.73 \cdot Hf_v \cdot Hf_i \cdot Hf_{pf} \tag{14}$$

where Hf_{pa} is the line feeder active power, Hf_v represents line feeder voltage, Hf_i is the line feeder current, and Hf_{pf} denotes the line feeder power factor. The function f_{14} determines the active power within the line feeder system. Monitoring active power in line feeders helps ensure efficient power delivery and prevent potential overloads in the distribution network.

15) LINE FEEDER REACTIVE POWER (LINE FEEDER MODULE)

$$Hf_{rp} = 1.73 \cdot Hf_v \cdot Hf_i \cdot (1 - Hf_{pf}) \tag{15}$$

where Hf_{rp} is the line feeder reactive power, and the function f_{15} determines the active power within the line feeder system. Managing reactive power in line feeders is essential to avoid power quality issues and maintain the efficiency of electrical transmission.

16) LINE FEEDER TOTAL POWER (LINE FEEDER MODULE)

$$Hf_{sp} = 1.73 \cdot Hf_v \cdot Hf_i \tag{16}$$

where Hf_{sp} is the line feeder total power, and the function f_{16} determines the active power within the line feeder system. Total power calculations for line feeders provide a holistic view of energy distribution and help optimize grid performance.

IV. ATTACK FRAMEWORK

Critical infrastructures are of great importance for the functioning of modern societies, and the energy systems within these infrastructures are of strategic importance in terms of economic stability, public safety, and national security. Therefore, protecting energy infrastructures in terms of cybersecurity and detecting potential threats to these systems is of vital importance. Any disruption, delay, or failure that may occur in such infrastructures can cause serious economic losses, damage prestige, disrupt public order, and even endanger human life. Therefore, while it is critical for energy infrastructures to undergo cybersecurity tests, tests performed directly on these systems can pose risks. In particular, tests performed on live systems can lead to large-scale interruptions and failures in the event of an error.

In this context, the use of digital twin technology offers a significant advantage to perform cybersecurity tests in a safer and more risk-free environment. Digital twin technology allows the creation of one-to-one copies of physical systems in a digital environment, allowing tests to be conducted without any impact on the real system. In this way, potential risks are minimized and the results of cybersecurity tests can be analyzed more accurately.

Within the scope of this study, an attack framework has been developed to perform cybersecurity tests on the digital twin model developed for energy infrastructures. This developed framework has been designed using the Python programming language and has the ability to simulate various cyberattack scenarios. Different types of cybersecurity attacks have been carried out through this attack framework and the potential effects of these attacks on the energy infrastructure have been examined in detail. The flow diagram showing the general operation of the framework is presented in Figure 5.

In the study, a cyberattack scenario targeting industrial communication protocols has been developed in the digital twin system. In this scenario, operations such as unauthorized reading of data, writing of data, and gathering information about the target system have been carried out. For this purpose, an attack framework has been created using the Python language, and attacks have been carried out against the Modbus TCP protocol through this framework.

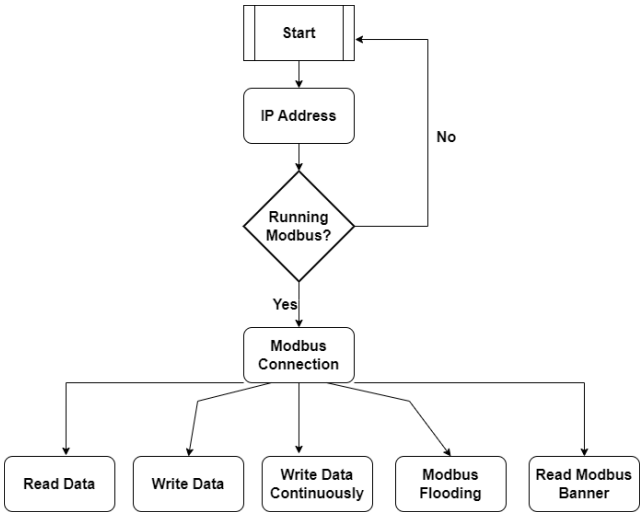


FIGURE 5. Attack framework flow.

The developed attack framework is designed to collect and intervene data from the HEPP via a Modbus TCP client. The framework includes functions such as reading and writing data, continuous data writing operations, flooding attacks on the Modbus protocol, and reading Modbus banners. In addition, the application is supported by a graphical user interface (GUI) that allows data to be saved in CSV format. Thanks to this graphical interface, attack processes can be managed in a user-friendly environment, and the results can be analyzed more systematically.

This study makes a significant contribution to the use of digital twin models in cybersecurity tests of energy infrastructures and shows an effective way to minimize the risks that can be created in real systems. This approach constitutes a model that is applicable not only for the energy sector but also for other critical infrastructures.

The Modbus Banner Reading Function of the Modbus Attack Framework, which has been prepared to collect preliminary information about the digital twin system of the HEPP, has been first checked for the existence of a Modbus TCP protocol on the target system. Then, the banner information has been read and information about the target system has been collected. The control of whether the Modbus TCP protocol is running on the target system is shown in Figure 6, and the images of the banner information being read with the Modbus Attack Framework and the Nmap tool [82] are given in Figures 7 and 8.

V. IMPLEMENTATION OF CYBER SECURITY TESTS

The importance of attack scenarios in the field of cybersecurity is increasing, and the simulation of these scenarios is becoming more comprehensive and effective with the

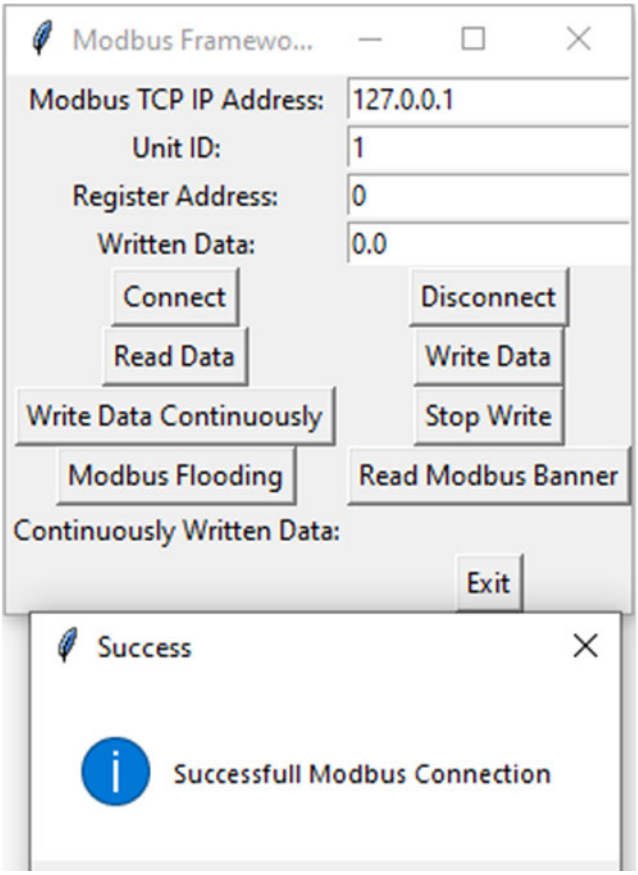


FIGURE 6. Control of the Modbus TCP protocol.

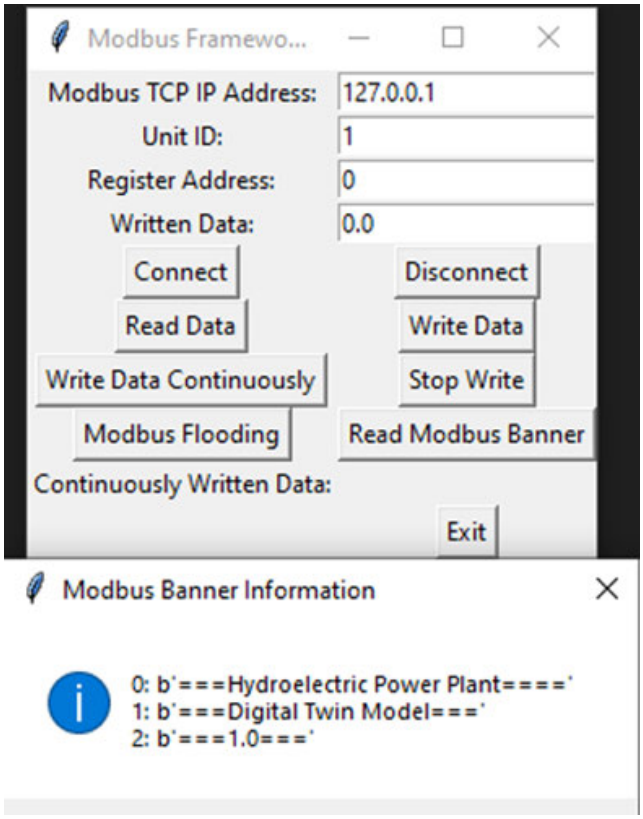


FIGURE 7. Reading Modbus banner information with Modbus attack framework.

```
Starting Nmap 7.92 ( https://nmap.org ) at 2024-10-01 12:02 Türkiye Standart Saati
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0010s latency).

PORT    STATE SERVICE
502/tcp open  modbus
| modbus-discover:
|   sid 0x1:
|     Slave ID data: ===Hydroelectric Power Plant====-===Digital Twin Model===-===1.0===\xFF
|_    Device identification: ===Hydroelectric Power Plant==== ===Digital Twin Model=== ===1.0===
```

FIGURE 8. Reading Modbus banner information with Nmap tool.

integration of digital twin technology. Digital twins create virtual copies of physical systems, allowing the collection and analysis of real-time data. In this way, the performance of defense mechanisms developed against attacks can be tested and the vulnerabilities of the system can be identified. With the use of digital twins, the cyber attack risks encountered in critical infrastructures such as energy, water, and transportation can be more thoroughly evaluated. For example, in the HEPP, the effects of possible attacks on the plant are modeled by simulating the water flow rate, dam gates, and generator performance. As a result of these modelings, the security vulnerabilities of the system are detected and more effective security protocols are developed against attacks. Digital twins also play a critical role in determining how operational processes are affected at the time of the attack, at which points vulnerabilities may occur, and in what way these vulnerabilities should be intervened in. In this way, critical operations in the plant can be tested in attack scenarios and prepared against real-world threats.

One of the important technologies in the use of digital twins in cybersecurity scenarios is Elasticsearch and related data analytics tools. Elasticsearch is a platform that is used for rapid indexing, storage, and searching of large data sets, in particular. The data used in the digital twin system is continuously collected through various sensors and devices, and the data is indexed on Elasticsearch. This allows real-time data to be analyzed during attack scenarios, and anomalies in different components of the system are quickly detected. For example, if the dam water level in a power plant drops abnormally, Elasticsearch quickly analyzes the data and integrates it into the digital twin model to detect possible failures or attack-related manipulations of the system. In this way, all data collected from physical sensors in the power plant are collected and analyzed in a single center, which supports the real-time decision mechanisms of the digital twin.

Another important technology used to model attack scenarios more effectively is Modbus and SCADA systems. Data collected from devices such as turbines, generators, and water pumps in Modbus enabled HEPPs are analyzed by the digital twin system and used in attack scenarios. SCADA systems are control systems used especially in industrial control systems, energy infrastructures, and water management. When these systems are integrated with the digital twin, the current state of each component of the physical infrastructure can

be monitored, and potential attacks can be simulated. For example, the effect of an attack on a SCADA system on the digital twin is simulated, and how the attack can damage the energy supply and what kind of defensive measures can be taken against such an attack are tested.

Digital twin technology can also be used together with visualization tools such as Kibana to provide a better understanding of the collected data and attack scenarios. Kibana visualizes the data indexed in Elasticsearch, and thus the effects of attack simulations can be analyzed graphically. These visualizations reveal where attacks are concentrated in the digital twin model and which parameters are at greater risk. In addition, the results of these analyzes are used to focus on what kind of countermeasures the digital twin can develop against attacks and how these countermeasures can be implemented in real-world systems. Visualizing the data obtained in attack scenarios allows both security teams and decision makers to better understand cyber threats and makes operational processes more secure.

A. CYBER SECURITY TEST SCENARIO

Water flow and nozzle aperture are two critical variables that directly affect the efficiency of energy production in HEPPs. While water flow expresses the amount of water that reaches the turbines of the plant, the nozzle aperture controls how quickly and intensively water reaches the turbines. The interaction of these two variables directly affects the efficiency of energy production processes. As the nozzle aperture narrows, the amount of water reaching the turbines decreases, and thus energy production decreases. However, a widening of the nozzle aperture allows water to reach the turbines more quickly and intensively, leading to increased energy production. However, this interaction may not be directly linear; because factors such as the mechanical structure of the system, pressure, and kinetic energy of the water also come into play. Therefore, keeping the nozzle aperture at the optimal level is of critical importance both in terms of maximizing energy production and preventing overloading of the turbines.

The relationship between water flow and nozzle aperture considered in this scenario is very important in understanding the dynamics of the energy production system. Although nozzle apertures allow plant operators to control water flow, water flow is the main determinant of energy production capacity. When the nozzle apertures are narrowed, the amount of water reaching the turbines decreases, directly reducing energy production. Such a narrowing can lead to serious losses in the energy production capacity and can cause the kinetic energy of water to not be used effectively. In contrast, widening the nozzle aperture allows the water to accelerate and a greater amount of water to reach the turbines, which increases energy production. However, if the nozzle aperture is widened too much, excessive pressure can be created on the turbines, which can lead to mechanical failure. Therefore, the optimal balance between these two variables is critical to maintaining energy production safely and efficiently.

In the cyber security test scenario, the digital twin model of the HEPP has been used to examine the change in water flow when nozzle 1 has been reduced from its initial aperture of 8.43 to 0. This change significantly reduced energy production in the plant and caused a serious decrease in the overall efficiency of the system. The decrease in water flow caused a dramatic decrease in the amount of water reaching the turbines, which in turn reduced the energy production capacity. This process has been monitored in real time via the digital twin and the effects of the manipulation of the nozzle aperture on the plant operations have been analyzed in detail. By reducing the nozzle aperture to 0, water has been almost completely prevented from reaching the turbines, and energy production came to a standstill. The data monitored on the HMI clearly revealed the effects of this manipulation on the water flow and thus possible risks to the operational processes of the plant have been observed.

In this scenario, a significant decrease has been observed not only in energy production but also in mechanical power as the nozzle aperture has been reduced to 0. The decrease in water flow directly affected the rotation speed of the turbines, leading to a decrease in mechanical power. The digital twin model detailed this relationship between water flow and mechanical power, revealing under what conditions the system could perform optimally. The decrease in mechanical power has been evaluated as a factor that could lead to unstable operation of the energy production system and potential failures. Thus, it has been understood how the system would be affected in the event of a possible attack or operational error in the power plant and how these effects would be reflected in the energy production process.

This scenario also shows how important a role digital twin technology plays in managing critical variables such as water flow and nozzle aperture. Thanks to the digital twin, the way water flow can be manipulated, the effects of these manipulations on energy production, and the security vulnerabilities of the system can be analyzed in detail. Simultaneous monitoring of water flow and nozzle aperture allows detection of anomalies in the energy production process and minimization of losses caused by these anomalies. Simulation of such scenarios is an important tool to increase operational safety and optimize energy production processes in HEPPs.

When calculating the water flow rate in the digital twin model, the mathematical formula between the nozzle apertures and the water flow rate is shown in (2) in Table 2. The water flow rate and nozzle apertures in the HMI before data manipulation are shown in Figure 9, and the visual of the frame prepared at the time of attack is shown in Figure 10.

During data manipulation, the nozzle 1 aperture, which has been 8.43, has been changed to 0 and, accordingly, the water flow rate has been changed according to (2). The relevant changes are shown on the HMI in Figure 11.

The changes that occurred after data manipulation have also been observed in Elasticsearch. The change in mechanical power in the digital twin model has occurred at the same

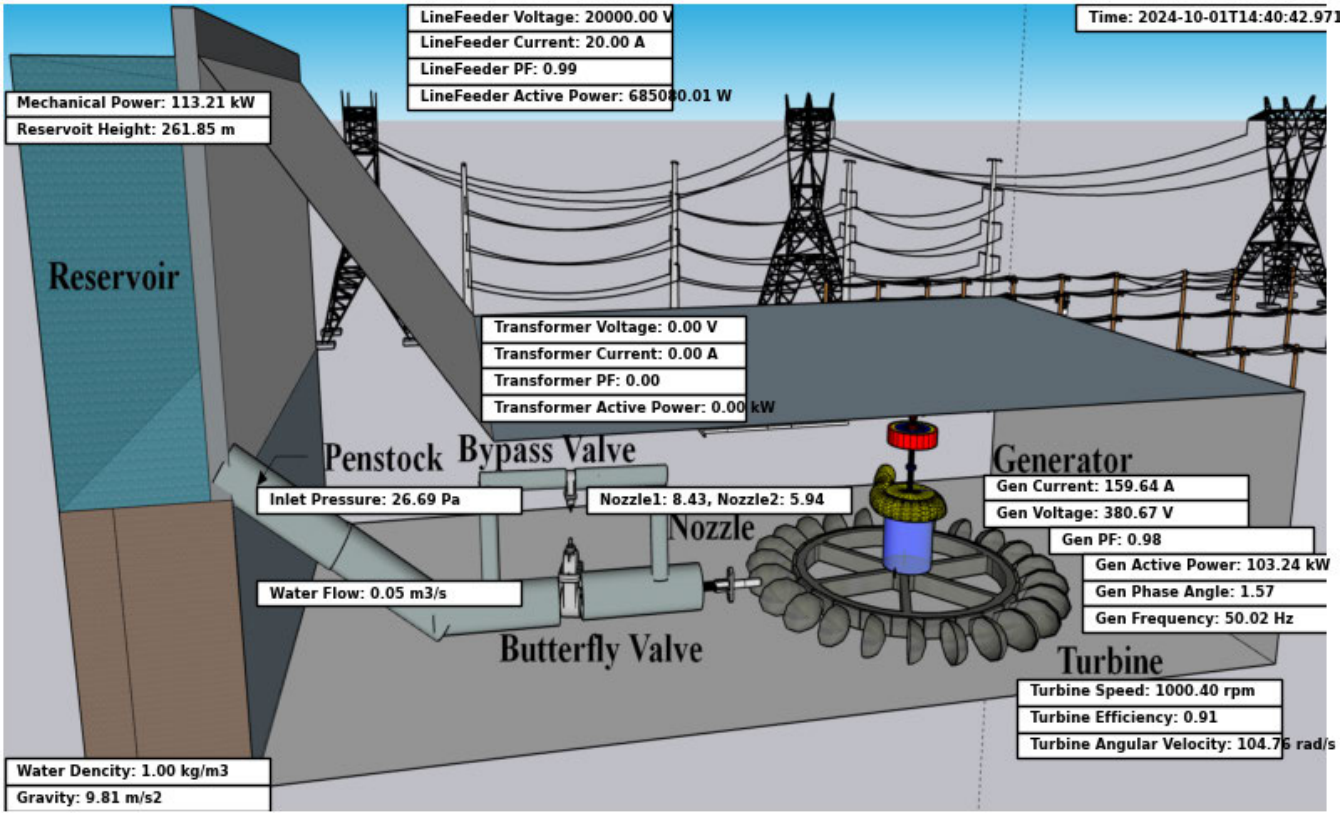


FIGURE 9. Water flow and nozzle apertures in HMI before data manipulation.

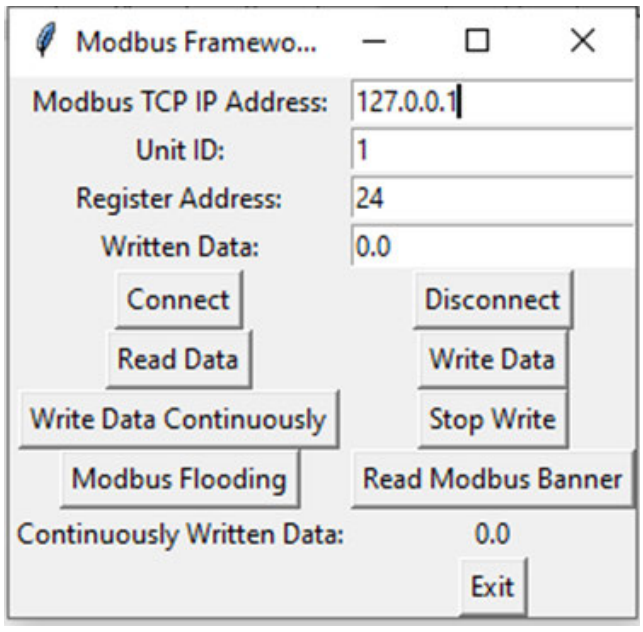


FIGURE 10. Modbus attack framework – nozzle 1 data manipulation.

rate depending on the change in water flow. The visual of the relevant changes is shown in Figure 12.

In the digital twin model, the water flow rate monitored in Elasticsearch is continuously compared with the actual water flow rate recorded from the power plant. This comparison allows for real-time tracking of discrepancies and validation of the model's accuracy against actual operational data. Figure 13 illustrates this comparison, providing a visual representation of how closely the digital twin aligns with the power plant's flow under various operating conditions. By analyzing the data, engineers can identify potential deviations, optimize model parameters, and ensure that the digital twin reflects the power plant performance with high fidelity. This approach improves predictive capabilities and operational insight, allowing more precise adjustments to be made to plant processes when necessary. Nozzle apertures are factors that directly affect the speed and amount of water that reaches the turbines. Reducing the nozzle aperture restricts the flow of water to the turbines and reduces the water flow rate. In this scenario, the nozzle 1 aperture has been reduced from 8.43 to 0, and as a result of this manipulation, the water flow rate decreased significantly. Data on HMI and Elasticsearch show that there has been a significant decrease in mechanical power immediately after this change.

Manipulations between water flow, and nozzle apertures directly affect the energy production capacity of the HEPP. Reducing the nozzle aperture restricts the amount of water reaching the turbines, causing a significant decrease in mechanical power. This decrease in mechanical power leads to serious losses in energy production and reduces the efficiency of the system. In a real system, such manipulation can cause operational problems and

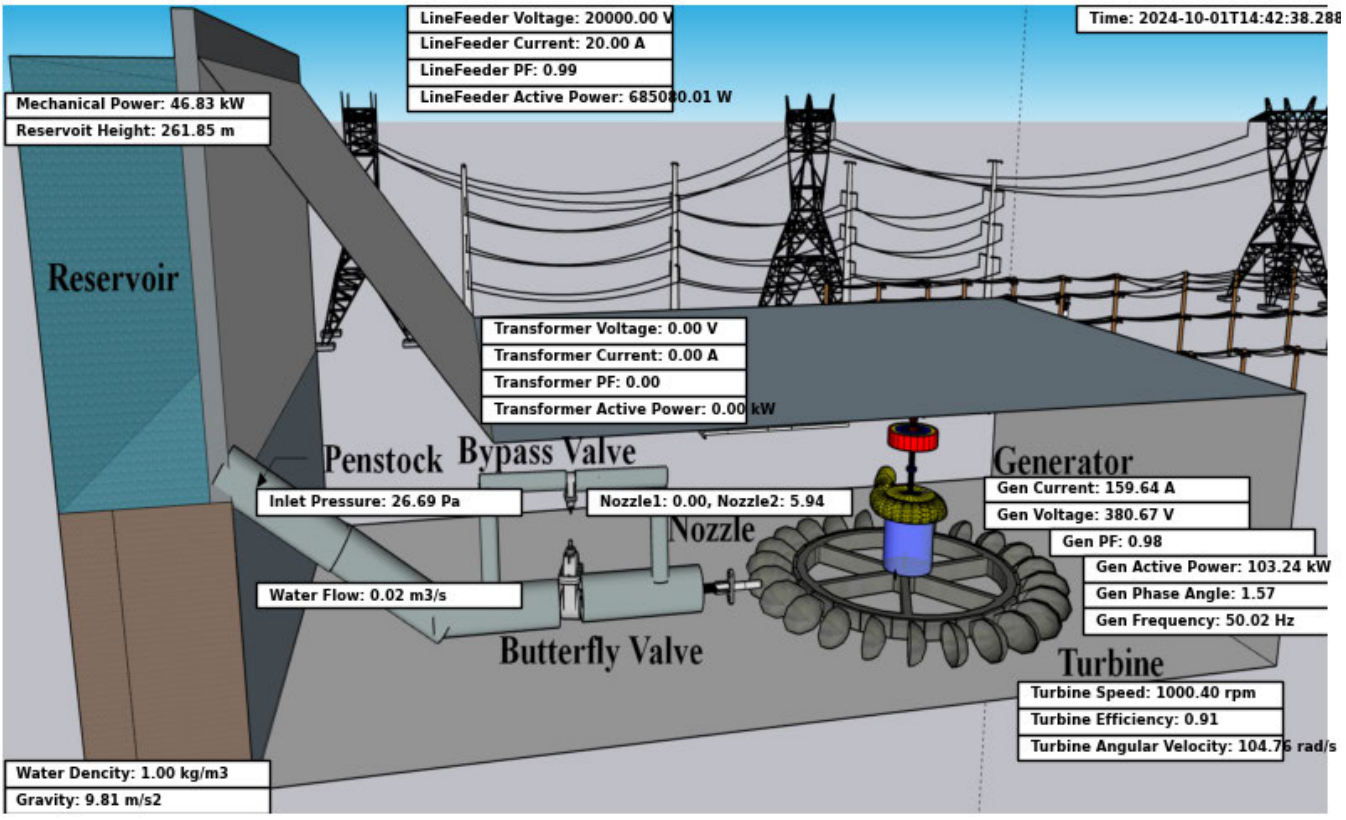


FIGURE 11. Water flow and nozzle apertures in HMI after data manipulation.

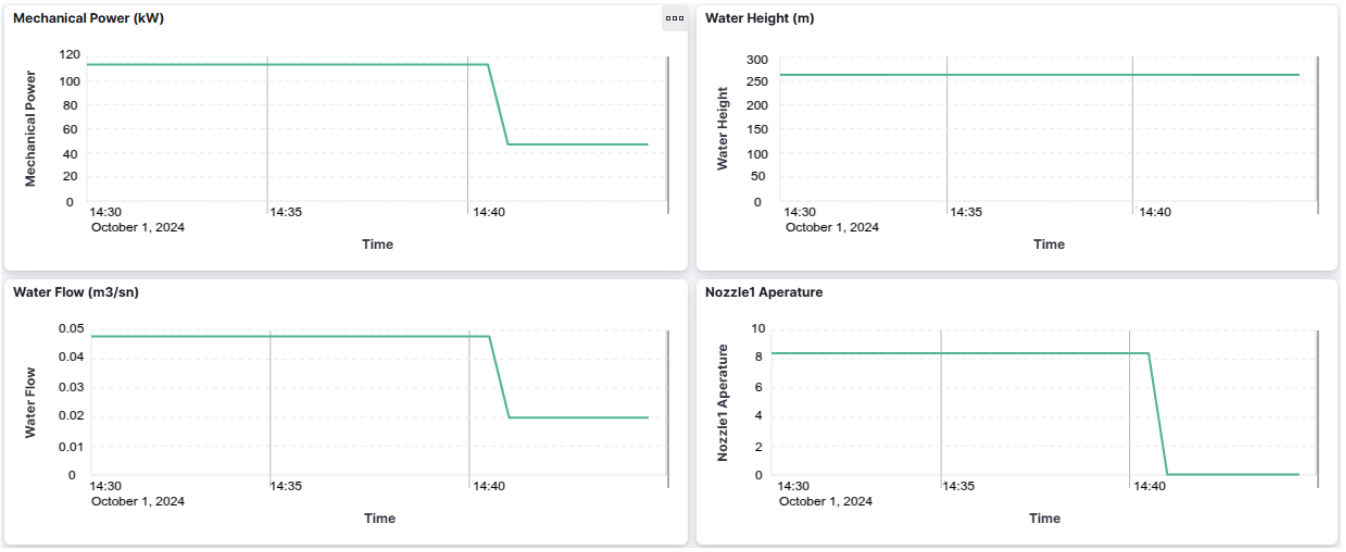


FIGURE 12. Change between water flow and mechanical power after data manipulation.

interruptions in energy supply during periods of high energy demand. Nozzle apertures and water flow rate manipulations affect not only energy production but also threaten the safety of the HEPP. Sudden reductions in water flow can cause turbines to operate erratically and potentially cause mechanical failures. These types of failures can increase maintenance costs in the long term and reduce overall system reliability. In addition, reduced water flow can cause turbines to overheat and other mechanical problems.

VI. ANOMALY DETECTION WITH XAI

XAI is an approach that aims to make the workings of machine learning and AI models transparent to human understanding. Traditional AI models are often referred to as ''black boxes'' because these models produce results

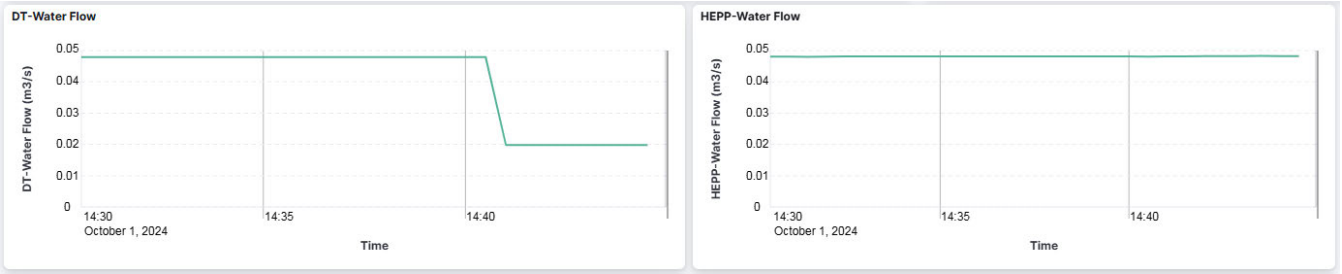


FIGURE 13. Comparison of water flow rate and power plant water flow rate values in the digital twin model.

using complex mathematical operations and algorithms, but users cannot easily understand how these processes work or how decisions are made [83]. This can create security and reliability issues, especially in critical areas such as healthcare, finance, and energy where it is important to know why and how AI-based decisions are made. To solve these problems, XAI offers methods and techniques that can explain how models work and why a particular decision has been made.

XAI techniques allow transparent examination of the behavior, decisions, and performance of machine learning models. This transparency allows users to better understand the model and increases the reliability of the results of AI systems. Especially for AI models used in high-risk areas, XAI techniques are of great importance in preventing wrong decisions or biased results.

XAI not only enables users to understand models, but also helps model developers develop more effective and reliable AI systems. Seeing which features of the model contribute to its results and how allows developers to identify deficiencies or biases in the model. It is also possible to understand when the model makes incorrect predictions and determine the reasons for these errors. This helps to continuously improve AI models and make them more reliable.

As a result, XAI provides the ability to explain the results of AI and machine learning models in a way that is compatible with human understanding. This not only increases the accuracy and reliability of AI applications, but also facilitates the widespread adoption of these systems. By bringing the complexity of AI systems closer to human understanding, XAI enables the development of safer and more effective AI applications, especially in critical decisions.

A. EXPERIMENTAL WORKS WITH XAI

In this work, an anomaly detection model is applied to the operational data of a HEPP using the SHAP technique, one of the XAI methods. This technique is used to evaluate the contribution of each feature that affects the output of the model and provides a better understanding by visualizing the results. As seen in Figure 14, SHAP shows how much positive or negative effect it has on each feature and the magnitude of this effect.

The SHAP method uses Shapley values based on game theory to explain the decisions of a machine learning model. Shapley values calculate the contribution of each feature to the decision made by the model. These values are used specially to understand the complex structure of the model and to analyze which features are more important in the model [83], [84], [85].

The calculation of SHAP values is carried out in three main steps:

Marginal Contributions of Features: The contribution of each feature to the model output is calculated independently of other features. For this, the change in the model output is observed before and after a feature is included.

Calculating Shapley Values: The Shapley value is calculated by averaging the contributions of the features before and after including a feature in all possible combinations. The Shapley value is a value that reflects the contribution of the features to the model in a more fair and balanced way.

Obtaining SHAP Values: SHAP values, which are a model-specific derivative of the Shapley values, are obtained as SHAP values for each data point and each feature by adapting to the prediction function of the machine learning model. The SHAP value of an observation i for a feature j is calculated as (17). Here is the LaTeX code for the equation:

$$\varnothing j(i) = \sum_{S \subseteq N(j)} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \quad (17)$$

where N is the set of all features, S represents the subset of features without feature j, $f(S)$ is the model's prediction when subset S is used as input, $f(S \cup \{j\}) - f(S)$ denotes the contribution of feature j to the model prediction, and $\frac{|S|!\cdot(|N|-|S|-1)!}{|N|!}$ provides balancing by accounting for the order of selected features. This formula takes the weighted average of the contribution of a feature to the model output in each combination.

To understand the overall behavior of the model, the average contribution of each feature can be calculated at all data points. This average value indicates the general importance of the features in model decisions.

To calculate the average SHAP value, the absolute SHAP value and its average are taken.

Obtaining the Absolute SHAP Values: First, the absolute SHAP values for each feature are taken. This allows us to

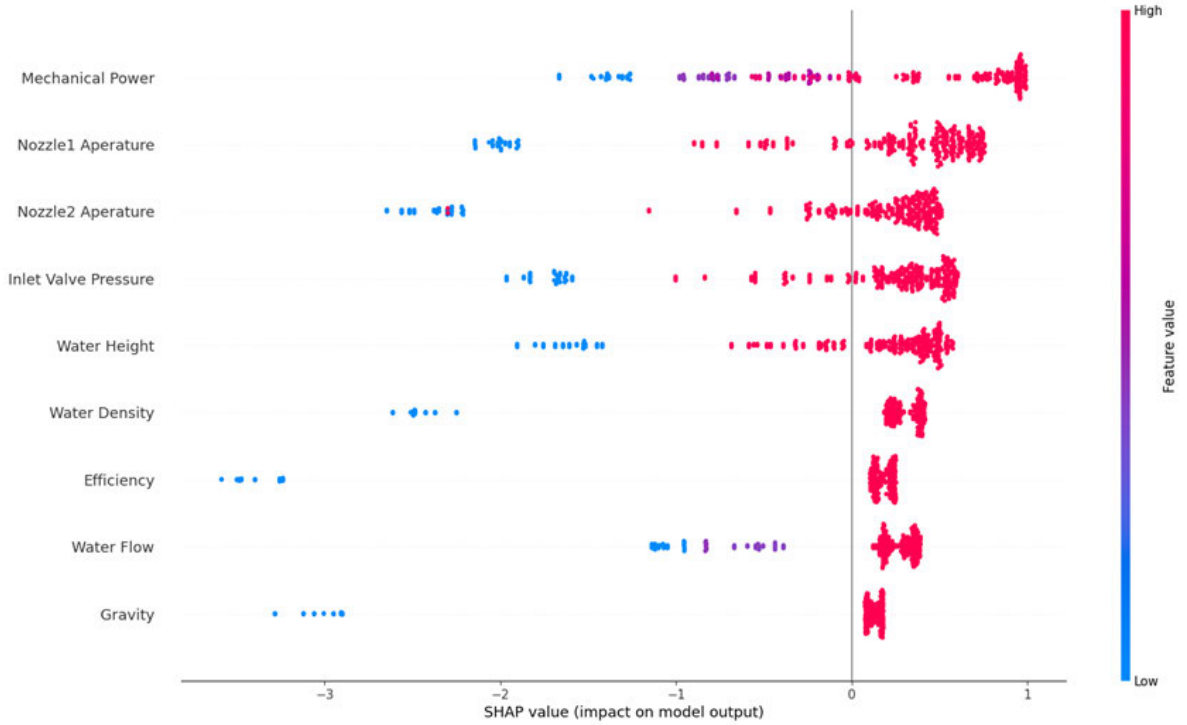


FIGURE 14. The effect of each variable in the model on the output using SHAP values.

measure the total contribution of the features to the model, regardless of whether the contribution is positive or negative.

Averaging: The absolute SHAP values are averaged for each feature. This shows which features the model is more dependent on in general.

Mathematically, the average SHAP value $\bar{\phi}_j$ for a feature j is calculated as shown in (18).

$$\bar{\phi}_j = \frac{1}{M} \sum_{i=1}^{M} |\phi_j(i)| \tag{18}$$

where M represents the total number of the data, and $\bar{\phi}_j(i)$ is the SHAP value of feature j for data point i.

This average value is an overall measure of the contribution of each feature in the model and helps understand which features the model considers more strongly.

B. EXPERIMENTAL RESULTS

First, different features of the HEPP have been brought together to train the model. Nozzle 1 aperture, nozzle 2 aperture, water height, mechanical power, and other characteristics represent various operational parameters of the plant. In particular, variables such as mechanical power, water height and nozzle apertures in the plant determine the energy production capacity and efficiency of the plant. Anomaly detection has been performed in the model using these data. The aim is to detect abnormal operational situations and determine potential risks that may occur in the plant in advance.

This data set consists of timestamps and various feature values that contain the details of each operational moment. With the SHAP technique, the reasons behind the decisions made by the model have been tried to make them understandable. During the training of the model, the decisions made from the data have been made explainable, and it has been observed which parameters had what effect on the system and which features have been dominant in anomalous situations.

Table 3 presents the SHAP values for various parameters of the HEPP, illustrating their relative contributions to the anomaly detection model. Among the parameters, nozzle 2 aperture exhibits the highest SHAP value (0.3642), indicating its dominant influence on the model predictions. This is closely followed by water flow (0.3575), water height (0.3261) and mechanical power (0.3439), which collectively underscore their critical roles in determining the operational state of the HEPP. In contrast, parameters such as water density (0.0018) and gravity (0.0001) contribute minimally, suggesting limited variability or impact on the anomalies detected. The notable importance of efficiency (0.3435) and inlet valve pressure (0.3232) further highlights the interconnected nature of operational parameters in influencing the system's performance. These findings emphasize the utility of SHAP analysis in providing transparent insights into parameter importance, allowing operators to prioritize monitoring and intervention strategies for key variables to enhance system reliability and security.

The effect of each characteristic according to the SHAP value is observed in Figure 14. The SHAP value is listed on the horizontal axis and the operational parameters of the plant

TABLE 3. SHAP values of the HEPP parameters.

Mechanical Power SHAP Value	Water Flow SHAP Value	Water Density SHAP Value	Nozzle 1 Aperture SHAP Value	Nozzle 2 Aperture SHAP Value	Gravity SHAP Value	Water Height SHAP Value	Efficiency SHAP Value	Inlet Valve Pressure SHAP Value
0.3439	0.3575	0.0018	0.2471	0.3642	0.0001	0.3261	0.3435	0.3232

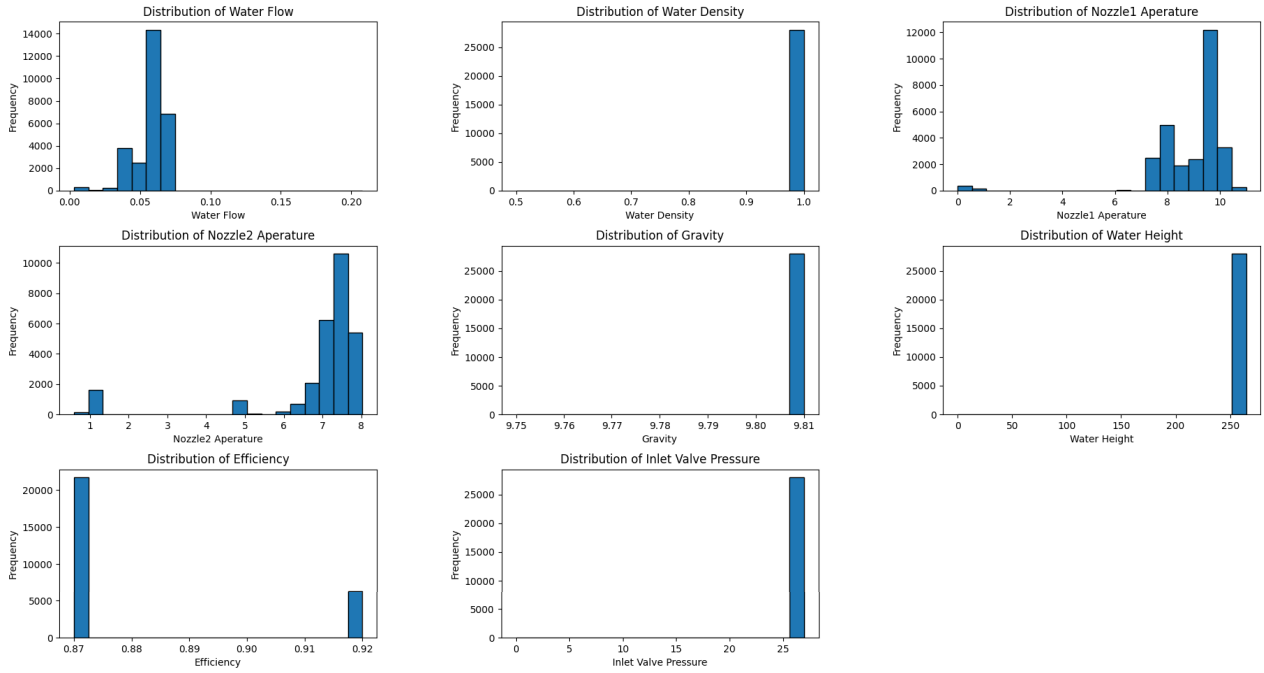


FIGURE 15. Distribution analysis of variables and their effects on SHAP values.

are listed on the vertical axis. The SHAP value shows the contribution of each feature to the model's output. Positive SHAP values affect the model's output upwards; negative SHAP values affect the output downward. The magnitude of the SHAP values shows the strength of the effect. For example, when the nozzle 2 aperture value is high (in red), it has a large positive effect on the model. Similarly, when the water height value is high (in red), it negatively affects the results of the model.

SHAP analysis used in this model not only explains the decisions of the model but also points out potential anomalies. For example, SHAP values outside the norm in features such as the nozzle 1 aperture and the nozzle 2 aperture indicate that there may be a problem in the system. Deviations observed in the data according to the timestamps may be a result of external interventions or operational errors in the system. In this context, the SHAP graph is quite useful to understand the reasons for the abnormal decisions made by the model.

In Figure 15, the distribution of each variable is observed in the HEPP data. Such distribution analyses play an important role in the calculation and interpretation of SHAP values, because the interactions of variables with each other and with the target variable during the model's learning process are directly related to the width and density of the data distribution.

For example, variables that show a narrower or more constant distribution may cause the model to attribute a lower importance to these variables. Especially features with a narrow distribution range, such as gravity, are not perceived by the model as a strong influencer as other variables, because the model evaluates the changes that such variables cause in the target variable as more limited. On the other hand, variables with a wide distribution, such as nozzle openings, may have a more pronounced effect in the SHAP analysis, as they provide more variation in the model. Thus, nozzle openings may show a greater effect in the SHAP values.

As seen in the Figure 15, the distribution width in the data set is reflected in the model's SHAP values. Variables with a wider distribution width create more variation in the model and therefore have a greater effect in the SHAP analysis. Such distribution analyses are important for the correct interpretation of SHAP values, because the way the model learns these variables is directly related to the distribution of these variables in the data set. Therefore, when interpreting SHAP values, it is important to consider the distribution properties of the variables in the data set.

TABLE 4. Comparison of literature and existing work on digital twin applications.

References	Comparison
[4], [5], [6], [7]	These studies address the use of digital twin technology in power systems for real-time monitoring, system optimization, and predictive maintenance. However, they do not incorporate Explainable AI (XAI) for anomaly detection or parameter interpretability. The current study advances the field by integrating XAI with digital twins, enabling transparent anomaly detection and improved decision-making in HEPPs
[11], [12], [13], [14]	Research on renewable energy systems highlights the use of digital twins for monitoring critical parameters and optimizing resource usage, such as wind and hydroelectric power. Unlike these works, the current study incorporates cybersecurity testing and anomaly detection using XAI techniques, offering a novel framework for resilience and operational security in HEPPs.
[17], [18], [19], [20]	Digital twins have been explored for cybersecurity applications, including vulnerability analysis and attack simulations. These studies, however, lack the integration of XAI methods like SHAP to prioritize critical system variables. The proposed study fills this gap by adding explainability to anomaly detection models, enhancing the practical usability of digital twins in HEPP cybersecurity.
[22], [23], [24], [25]	Applications of digital twins in smart grids focus on energy distribution, renewable integration, and system optimization. While these studies address general smart grid applications, the current study uniquely tailors digital twins to HEPP-specific operational requirements, integrating XAI for enhanced anomaly detection and cybersecurity.
[26], [28], [30], [31]	Digital twins have been applied in industrial sectors for predictive maintenance, operational optimization, and SCADA integration. However, these studies lack focus on integrating advanced XAI techniques. The current research extends these concepts by making anomaly detection models interpretable and applicable to HEPPs, providing actionable insights for operators.
[52], [53], [57], [58]	Digital twin applications in smart cities have been used for urban energy management and resource sustainability. Unlike these studies, the current work focuses on energy-specific infrastructures, addressing HEPP cybersecurity and resilience while incorporating XAI for transparency and anomaly detection.
[66], [67], [68], [71], [72]	Studies on XAI techniques, such as SHAP and LIME, emphasize improving transparency in anomaly detection models. However, they do not specifically address HEPP applications. The current study uniquely applies XAI techniques to hydroelectric systems, offering insights into parameter interactions and enhancing operational transparency.

It should also be emphasized that the findings obtained with the SHAP technique are important not only to increase the accuracy of the model, but also to increase the understandability of the model. Machine learning models are often considered 'black boxes' and are difficult to understand for users. However, XAI methods such as SHAP make it easier to understand how models work and to what extent which features affect the model results. Thus, model outputs are made more transparent, and their reliability is increased.

In the data, anomalies observed in parameters such as water height and mechanical power can be clearly seen according to the SHAP values. Such deviations indicate potential risks that may directly affect the power plant's energy production capacity. For example, low SHAP values of the nozzle 1 aperture may indicate situations where the nozzle is closed or does not perform its function. At the same time, abnormal values in mechanical power may indicate that the mechanical power is not at expected levels.

As a result, the SHAP technique used in this study stands out as an effective method to detect anomalies in the operational data of the HEPP and to explain the model's decisions. Especially, deviations observed in critical operational parameters provide important information for plant management and allow for the early detection of potential failures.

The 4 provides a comparative analysis of the current study with existing works in the literature, highlighting key advancements and contributions. References are categorized based on their thematic focus, such as power systems, renewable energy, cybersecurity, smart grids, industrial applications, and anomaly detection with XAI. Although similarities emphasize the shared objectives of using digital twin technology for real-time monitoring, system optimization, and predictive maintenance, notable differences are observed in the integration of Explainable AI (XAI) and its application to HEPPs. The current study addresses critical gaps by introducing a novel framework that combines digital twins with XAI techniques, enabling transparent anomaly detection and enhanced cybersecurity measures. This approach not only contributes to operational efficiency but also advances the resilience of HEPPs against emerging threats, distinguishing it from prior works. The table effectively summarizes the contributions of the study in relation to the state of the art, demonstrating its relevance and innovation within the field.

VII. CONCLUSION

This study presents a comprehensive framework that integrates DT technology and XAI for anomaly detection and cybersecurity in HEPPs. By creating a virtual representation of a real-world HEPP, the DT enables real-time monitoring, optimization, and risk-free cybersecurity testing. Key findings demonstrate that the DT model effectively identifies vulnerabilities, such as those in the Modbus TCP protocol, and provides a safe testing environment for cyberattack simulations. Integration with XAI techniques, particularly SHAP analysis, enhances the transparency and interpretability of anomaly detection models, revealing critical operational parameters while reducing the focus on less influential ones, such as gravity and water density.

The novelty of the study lies in the combination of DT and XAI to provide operational insight and resilience against cyber threats. This approach not only optimizes system performance, but also advances the understanding of complex interactions among HEPP parameters. However, the study acknowledges limitations, including the challenge of fully replicating real-world dynamics in the DT model and the

potential computational cost of large-scale implementations. Future research could focus on refining these models for broader applications in critical energy infrastructures and incorporating additional real-time data streams to improve predictive accuracy.

The contributions of this work extend to both academia and industry by offering actionable insights for sustainable and secure energy production. The proposed framework provides a scalable solution that improves HEPP efficiency, increases resilience against cyber threats, and serves as a foundation for the integration of advanced digital tools into critical infrastructure systems.

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